Iterative Methods

As discussed earlier, iterative methods are based on the idea of successive approximations. We start with an initial approximation to the solution vector $\mathbf{x} = \mathbf{x}_0$, to solve the system of equations $\mathbf{Ax} = \mathbf{b}$, and obtain a sequence of approximate vectors $\mathbf{x}_0, \mathbf{x}_1, \ldots, \mathbf{x}_k, \ldots$, which in the limit as $k \to \infty$, converges to the exact solution vector $\mathbf{x} = \mathbf{A}^{-1}\mathbf{b}$. A general linear iterative method for the solution of the system of equations $\mathbf{Ax} = \mathbf{b}$, can be written in matrix form as

$$\mathbf{x}^{(k+1)} = \mathbf{Hx}^{(k)} + \mathbf{c}, \quad k = 0, 1, 2, \ldots \tag{1}$$

where $\mathbf{x}^{(k+1)}$ and $\mathbf{x}^{(k)}$ are the approximations for $\mathbf{x}$ at the $(k + 1)$th and kth iterations respectively. $\mathbf{H}$ is called the iteration matrix, which depends on $\mathbf{A}$ and $\mathbf{c}$ is a column vector, which depends on $\mathbf{A}$ and $\mathbf{b}$.

When to stop the iteration We stop the iteration procedure when the magnitudes of the differences between the two successive iterates of all the variables are smaller than a given accuracy or *error tolerance* or an error bound ε, that is,

$$\left| x_i^{(k+1)} - x_i^{(k)} \right| \leq \varepsilon, \quad \text{for all } i. \tag{2}$$

For example, if we require two decimal places of accuracy, then we iterate until $\left| x_i^{(k+1)} - x_i^{(k)} \right| < 0.005$, for all i. If we require three decimal places of accuracy, then we iterate until $\left| x_i^{(k+1)} - x_i^{(k)} \right| < 0.0005$, for all i.

Convergence property of an iterative method depends on the iteration matrix **H.**

Now, we derive two iterative methods for the solution of the system of algebraic equations

$$\begin{aligned} a_{11}x_1 + a_{12}x_2 + a_{13}x_3 &= b_1 \\ a_{21}x_1 + a_{22}x_2 + a_{23}x_3 &= b_2 \\ a_{31}x_1 + a_{32}x_2 + a_{33}x_3 &= b_3 \end{aligned} \tag{3}$$

Gauss-Jacobi Iteration Method

Sometimes, the method is called *Jacobi method*. We assume that the pivots $a_{ii} \neq 0$, for all i. Write the equations as

$$a_{11}x_1 = b_1 - (a_{12}x_2 + a_{13}x_3)$$
$$a_{22}x_2 = b_2 - (a_{21}x_1 + a_{23}x_3)$$
$$a_{33}x_3 = b_3 - (a_{31}x_1 + a_{32}x_2)$$

The Jacobi iteration method is defined as

$$x_1^{(k+1)} = \frac{1}{a_{11}}[b_1 - (a_{12}x_2^{(k)} + a_{13}x_3^{(k)})]$$

$$x_2^{(k+1)} = \frac{1}{a_{22}}[b_2 - (a_{21}x_1^{(k)} + a_{23}x_3^{(k)})]$$

$$x_3^{(k+1)} = \frac{1}{a_{33}}[b_3 - (a_{31}x_1^{(k)} + a_{32}x_2^{(k)})], \quad k = 0, 1, 2, \ldots \qquad (4)$$

Since, we replace the complete vector $\mathbf{x}^{(k)}$ in the right hand side of (4) at the end of each iteration, this method is also called the *method of simultaneous displacement.*

Remark *A sufficient condition for convergence of the Jacobi method is that the system of equations is diagonally dominant*, that is, the coefficient matrix **A** is diagonally dominant. We can verify that $| a_{ii} | \geq \sum_{j=1, i \neq j}^{n} |a_{ij}|$. This implies that convergence may be obtained even if the system is not diagonally dominant. If the system is not diagonally dominant, we may exchange the equations, if possible, such that the new system is diagonally dominant and convergence is guaranteed. However, such manual verification or exchange of equations may not be possible for large systems that we obtain in application problems. The necessary and sufficient condition for convergence is that the spectral radius of the iteration matrix **H** is less than one unit, that is, $\rho(\mathbf{H}) < 1$, where $\rho(\mathbf{H})$ is the largest eigen value in magnitude of **H.** Testing of this condition is beyond the scope of the syllabus.

Remark *How do we find the initial approximations to start the iteration*? If the system is diagonally dominant, then the iteration converges for any initial solution vector. If no suitable approximation is available, we can choose $\mathbf{x} = \mathbf{0}$, that is $x_i = 0$ for all i. Then, the initial approximation becomes $x_i = b_i / a_{ii}$, for all i.

Example *Solve the system of equations*

$$4x_1 + x_2 + x_3 = 2$$
$$x_1 + 5x_2 + 2x_3 = -6$$
$$x_1 + 2x_2 + 3x_3 = -4$$

using the Jacobi iteration method. Use the initial approximations as

(*i*) $x_i = 0,\ i = 1, 2, 3,$ (*ii*) $x_1 = 0.5,\ x_2 = -0.5,\ x_3 = -0.5.$

Perform five iterations in each case.

Solution Note that the given system is diagonally dominant. Jacobi method gives the iterations as

$$x_1^{(k+1)} = 0.25\ [2 - (x_2^{(k)} + x_3^{(k)})]$$
$$x_2^{(k+1)} = 0.2\ [-6 - (x_1^{(k)} + 2x_3^{(k)})]$$
$$x_3^{(k+1)} = 0.33333\ [-4 - (x_1^{(k)} + 2x_2^{(k)})], \quad k = 0, 1, \ldots$$

We have the following results.

(*i*) $x_1^{(0)} = 0,\ x_2^{(0)} = 0,\ x_3^{(0)} = 0.$

First iteration

$$x_1^{(1)} = 0.25\ [2 - (x_2^{(0)} + x_3^{(0)})] = 0.5,$$

$$x_2^{(1)} = 0.2\ [-6 - (x_1^{(0)} + 2x_3^{(0)})] = -1.2,$$

$$x_3^{(1)} = 0.33333\ [-4 - (x_1^{(0)} + 2x_2^{(0)})] = -1.33333.$$

Second iteration

$$x_1^{(2)} = 0.25\ [2 - (x_2^{(1)} + x_3^{(1)})] = 0.25\ [2 - (-1.2 - 1.33333)] = 1.13333,$$

$$x_2^{(2)} = 0.2\ [-6 - (x_1^{(1)} + 2x_3^{(1)})] = 0.2\ [-6 - (0.5 + 2(-1.33333))] = -0.76668,$$

$$x_3^{(2)} = 0.33333\ [-4 - (x_1^{(1)} + 2x_2^{(1)})] = 0.33333\ [-4 - (0.5 + 2(-1.2))] = -0.7.$$

Third iteration

$$x_1^{(3)} = 0.25\ [2 - (x_2^{(2)} + x_3^{(2)})] = 0.25\ [2 - (-0.76668 - 0.7)] = 0.86667,$$

$$x_2^{(3)} = 0.2\ [-6 - (x_1^{(2)} + 2x_3^{(2)})] = 0.2\ [-6 - (1.13333 + 2(-0.7))] = -1.14667,$$

$$x_3^{(3)} = 0.33333\ [-4 - (x_1^{(2)} + 2x_2^{(2)})]$$

$$= 0.33333\ [-4 - (1.13333 + 2(-0.76668))] = -1.19998.$$

Fourth iteration

$$x_1^{(4)} = 0.25\ [2 - (x_2^{(3)} + x_3^{(3)})] = 0.25\ [2 - (-\ 1.14667 - 1.19999)] = 1.08666,$$

$$x_2^{(4)} = 0.2\ [-\ 6 - (x_1^{(3)} + 2x_3^{(3)})] = 0.2\ [-\ 6 - (0.86667 + 2(-\ 1.19998))] = -\ 0.89334,$$

$$x_3^{(4)} = 0.33333\ [-\ 4 - (x_1^{(3)} + 2x_2^{(3)})]$$

$$= 0.33333\ [-\ 4 - (0.86667 + 2(-\ 1.14667))] = -\ 0.85777.$$

Fifth iteration

$$x_1^{(5)} = 0.25\ [2 - (x_2^{(4)} + x_3^{(4)})] = 0.25\ [2 - (-\ 0.89334 - 0.85777)] = 0.93778,$$

$$x_2^{(5)} = 0.2\ [-\ 6 - (x_1^{(4)} + 2x_3^{(4)})] = 0.2\ [-\ 6 - (1.08666 + 2(-\ 0.85777))] = -\ 1.07422,$$

$$x_3^{(5)} = 0.33333\ [-\ 4 - (x_1^{(4)} + 2x_2^{(4)})]$$

$$= 0.33333\ [-\ 4 - (1.08666 + 2(-\ 0.89334))] = -\ 1.09998.$$

It is interesting to note that the iterations oscillate and converge to the exact solution $x_1 = 1.0$, $x_2 = -\ 1$, $x_3 = -\ 1.0$.

(*ii*) $x_1^{(0)} = 0.5,\ x_2^{(0)} = -0.5,\ x_3^{(0)} = -0.5.$

First iteration

$$x_1^{(1)} = 0.25\ [2 - (x_2^{(0)} + x_3^{(0)})] = 0.25\ [2 - (-0.5 - 0.5)] = 0.75,$$

$$x_2^{(1)} = 0.2\ [-6 - (x_1^{(0)} + 2x_3^{(0)})] = 0.2\ [-6 - (0.5 + 2(-0.5))] = -1.1,$$

$$x_3^{(1)} = 0.33333\ [-4 - (x_1^{(0)} + 2x_2^{(0)})] = 0.33333\ [-4 - (0.5 + 2(-0.5))] = -1.16667.$$

Second iteration

$$x_1^{(2)} = 0.25\ [2 - (x_2^{(1)} + x_3^{(1)})] = 0.25\ [2 - (-1.1 - 1.16667)] = 1.06667,$$

$$x_2^{(2)} = 0.2\ [-6 - (x_1^{(1)} + 2x_3^{(1)})] = 0.2\ [-6 - (0.75 + 2(-1.16667))] = -0.88333,$$

$$x_3^{(2)} = 0.33333\ [-4 - (x_1^{(1)} + 2x_2^{(1)})] = 0.33333\ [-4 - (0.75 + 2(-1.1))] = -0.84999.$$

Third iteration

$$x_1^{(3)} = 0.25\ [2 - (x_2^{(2)} + x_3^{(2)})] = 0.25\ [2 - (-0.88333 - 0.84999)] = 0.93333,$$

$$x_2^{(3)} = 0.2\ [-6 - (x_1^{(2)} + 2x_3^{(2)})] = 0.2\ [-6 - (1.06667 + 2(-0.84999))] = -1.07334,$$

$$x_3^{(3)} = 0.33333\ [-4 - (x_1^{(2)} + 2x_2^{(2)})]$$

$$= 0.33333\ [-4 - (1.06667 + 2(-0.88333))] = -1.09999.$$

Fourth iteration

$$x_1^{(4)} = 0.25\ [2 - (x_2^{(3)} + x_3^{(3)})] = 0.25\ [2 - (-1.07334 - 1.09999)] = 1.04333,$$

$$x_2^{(4)} = 0.2\ [-6 - (x_1^{(3)} + 2x_3^{(3)})] = 0.2\ [-6 - (0.93333 + 2(-1.09999))] = -0.94667,$$

$$x_3^{(4)} = 0.33333\ [-4 - (x_1^{(3)} + 2x_2^{(3)})]$$

$$= 0.33333\ [-4 - (0.93333 + 2(-1.07334))] = -0.92887.$$

Fifth iteration

$$x_1^{(5)} = 0.25\ [2 - (x_2^{(4)} + x_3^{(4)})] = 0.25\ [2 - (-0.94667 - 0.92887)] = 0.96889,$$

$$x_2^{(5)} = 0.2\ [-6 - (x_1^{(4)} + 2x_3^{(4)})] = 0.2\ [-6 - (1.04333 + 2(-0.92887))] = -1.03712,$$

$$x_3^{(5)} = 0.33333\ [-4 - (x_1^{(4)} + 2x_2^{(4)})]$$

$$= 0.33333\ [-4 - (1.04333 + 2(-0.94667))] = -1.04999.$$

Example *Solve the system of equations*

$$26x_1 + 2x_2 + 2x_3 = 12.6$$

$$3x_1 + 27x_2 + x_3 = -14.3$$

$$2x_1 + 3x_2 + 17x_3 = 6.0$$

using the Jacobi iteration method. Obtain the result correct to three decimal places.

Solution The given system of equations is strongly diagonally dominant. Hence, we can expect faster convergence. Jacobi method gives the iterations as

$$x_1^{(k+1)} = [12.6 - (2x_2^{(k)} + 2x_3^{(k)})]/26$$

$$x_2^{(k+1)} = [-14.3 - (3x_1^{(k)} + x_3^{(k)})]/27$$

$$x_3^{(k+1)} = [6.0 - (2x_1^{(k)} + 3x_2^{(k)})]/17 \qquad k = 0, 1, \ldots$$

Choose the initial approximation as $x_1^{(0)} = 0$, $x_2^{(0)} = 0$, $x_3^{(0)} = 0$. We obtain the following results.

First iteration

$$x_1^{(1)} = \frac{1}{26}\,[12.6 - (2x_2^{(0)} + 2x_3^{(0)})] = \frac{1}{26}\,[12.6] = 0.48462,$$

$$x_2^{(1)} = \frac{1}{27}\,[-14.3 - (3x_1^{(0)} + x_3^{(0)})] = \frac{1}{27}\,[-14.3] = -0.52963,$$

$$x_3^{(1)} = \frac{1}{17}\,[6.0 - (2x_1^{(0)} + 3x_2^{(0)})] = \frac{1}{17}\,[6.0] = 0.35294.$$

Second iteration

$$x_1^{(2)} = \frac{1}{26}\,[12.6 - (2x_2^{(1)} + 2x_3^{(1)})] = \frac{1}{26}\,[12.6 - 2\,(-0.52963 + 0.35294)] = 0.49821,$$

$$x_2^{(2)} = \frac{1}{27}\,[-14.3 - (3x_1^{(1)} + x_3^{(1)})] = \frac{1}{27}\,[-14.3 - (3(0.48462) + 0.35294)] = -0.59655,$$

$$x_3^{(2)} = \frac{1}{17}\,[-6.0 - (2x_1^{(1)} + 3x_2^{(1)})] = \frac{1}{17}\,[6.0 - (2(0.48462) + 3(-0.52963))] = 0.38939.$$

Third iteration

$$x_1^{(3)} = \frac{1}{26}\,[12.6 - (2x_2^{(2)} + 2x_3^{(2)})] = \frac{1}{26}\,[12.6 - 2(-0.59655 + 0.38939)] = 0.50006,$$

$$x_2^{(3)} = \frac{1}{27}\,[-14.3 - (3x_1^{(2)} + x_3^{(2)})] = \frac{1}{27}\,[-14.3 - (3(0.49821) + 0.38939)] = -0.59941,$$

$$x_3^{(3)} = \frac{1}{17}\,[-6.0 - (2x_1^{(2)} + 3x_2^{(2)})] = \frac{1}{17}\,[6.0 - (2(0.49821) + 3(-0.59655))] = 0.39960.$$

Fourth iteration

$$x_1^{(4)} = \frac{1}{26}\,[12.6 - (2x_2^{(3)} + 2x_3^{(3)})] = \frac{1}{26}\,[12.6 - 2(-0.59941 + 0.39960)] = 0.50000,$$

$$x_2^{(4)} = \frac{1}{27}\,[-14.3 - (3x_1^{(3)} + x_3^{(3)})] = \frac{1}{27}\,[-14.3 - (3(0.50006) + 0.39960)] = -0.59999,$$

$$x_3^{(4)} = \frac{1}{17}\,[-6.0 - (2x_1^{(3)} + 3x_2^{(3)})] = \frac{1}{17}\,[6.0 - (2(0.50006) + 3(-0.59941))] = 0.39989.$$

We find $\left| x_1^{(4)} - x_1^{(3)} \right| = |\,0.5 - 0.50006\,| = 0.00006,$

$$\left| x_2^{(4)} - x_2^{(3)} \right| = |\,-0.59999 + 0.59941\,| = 0.00058,$$

$$\left| x_3^{(4)} - x_3^{(3)} \right| = |\,0.39989 - 0.39960\,| = 0.00029.$$

The required solution is $x_1 = 0.5, x_2 = -0.59999, x_3 = 0.39989.$